B.Tech. Final Year Project (2019-2023)

Broad Area of Project

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Group No: 22

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Introduction to Broad Area of your Research:
Lattice-based cryptography is a rapidly developing area of research that has the potential to provide strong security for a wide range of applications. Lattices are mathematical structures that can be used to construct cryptographic schemes such as fully homomorphic encryption, identity-based encryption, and lattice-based signatures.

However, the security of these schemes can be compromised if an attacker is able to find short vectors in the underlying lattice. Lattice reduction algorithms are a crucial tool in the field of cryptanalysis, as they allow researchers to find short vectors in a lattice and use them to break certain types of lattice-based cryptographic schemes.

Application/ Why it is Important?
The development of efficient and secure lattice reduction algorithms is important for several reasons. Firstly, lattice-based cryptographic schemes have the potential to provide strong security for a wide range of applications, such as secure communication, secure multiparty computation, and digital signatures.

However, the security of these schemes relies on the difficulty of finding short vectors in the underlying lattice. Improving the efficiency and security of lattice reduction algorithms can help to ensure that these schemes are secure against attack. Secondly, the growing use of quantum computers has raised concerns about the security of traditional cryptographic techniques such as RSA and ECC.

Lattice-based cryptography has been proposed as a potential solution to this problem, and efficient and secure lattice reduction algorithms are necessary for breaking these schemes.

Design Issues / Requirements /Need for the further study:

There are several design issues that need to be considered when developing lattice reduction algorithms. Efficiency is a key concern, as these algorithms are often used as subroutines in other cryptographic schemes, and the overall performance of these schemes depends on the efficiency of the lattice reduction algorithm.

In addition, security is also a major concern, as an attacker who is able to break the lattice reduction algorithm could potentially use it to break the underlying cryptographic scheme. Finally, there is a need for further study to understand the limitations and trade-offs of different lattice reduction algorithms, and to identify the most promising approaches for improving their efficiency and security.

Issue that will be addressed by your Project:

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